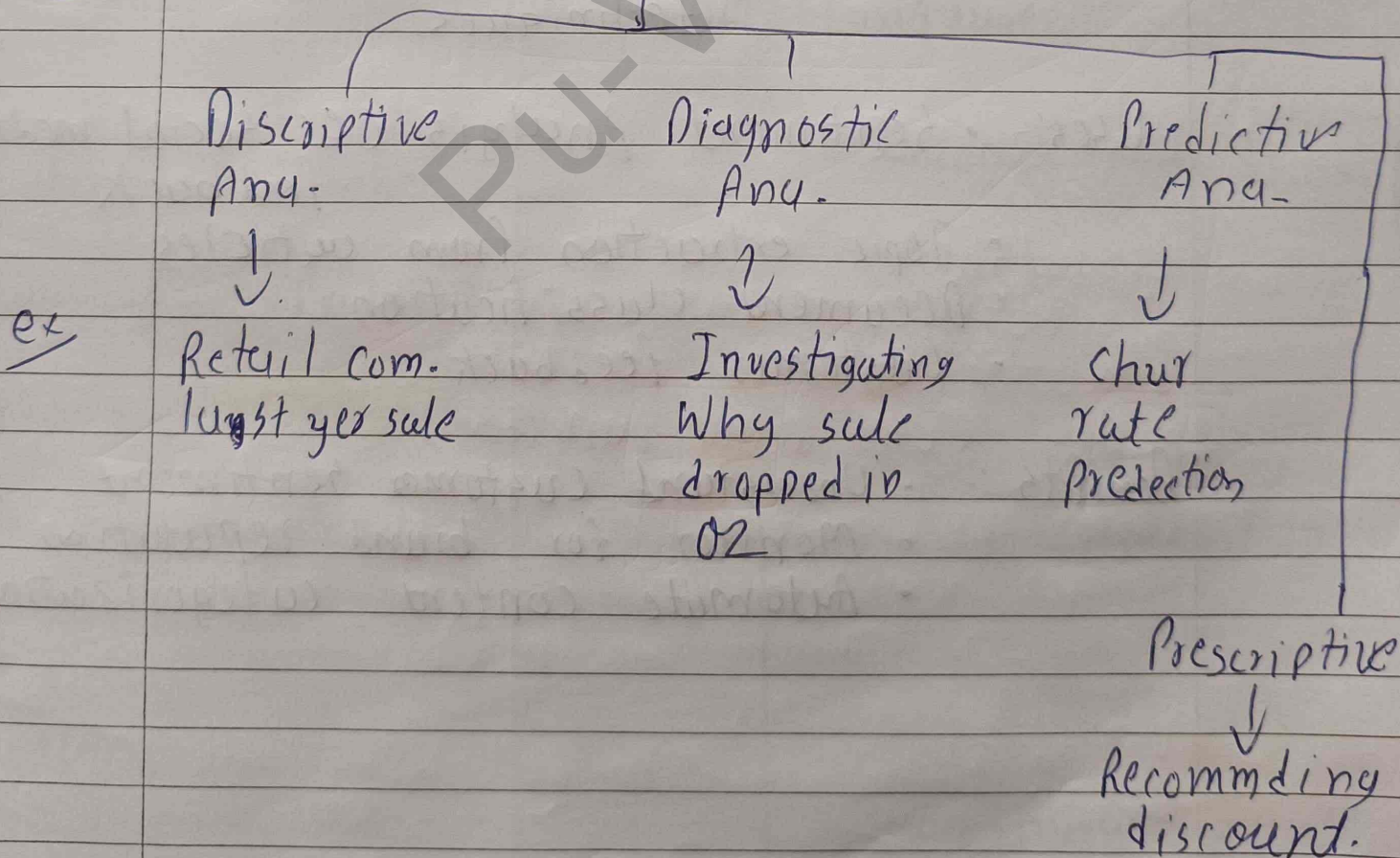


① Difference between Reporting & Analytics

feature	Reporting	Analysis
focus	Descriptive	Prescriptive
Output	Dashboard, chart	Model, insight
Tools	PowerBi, SQL	ML Model, R, Py
Time-fe.	Historical	Historical + future

② Type of Analytics



//_

③ Steps to develop An Analytics Team in Org -

- Define Goals
- Hire key role
- Infrastructure setup
- Data governance
- Process Design
- Training

④ Text analytics:- This is a process of extracting useful insights & pattern from unstructured text data using ML techniques.

Uses:-

- Sentiment Analysis (ex social media feedback)
- Topic extraction from articles
- Document classification
- Customer feedback

Benefits :-

- Understand customer sentiment
- Monitor for brand reputation
- Automate content categorization

Q

What is Apache pig how it differs from Traditional Database

Apache pig:- it is high-level platform for creating MapReduce program for big-data processing.

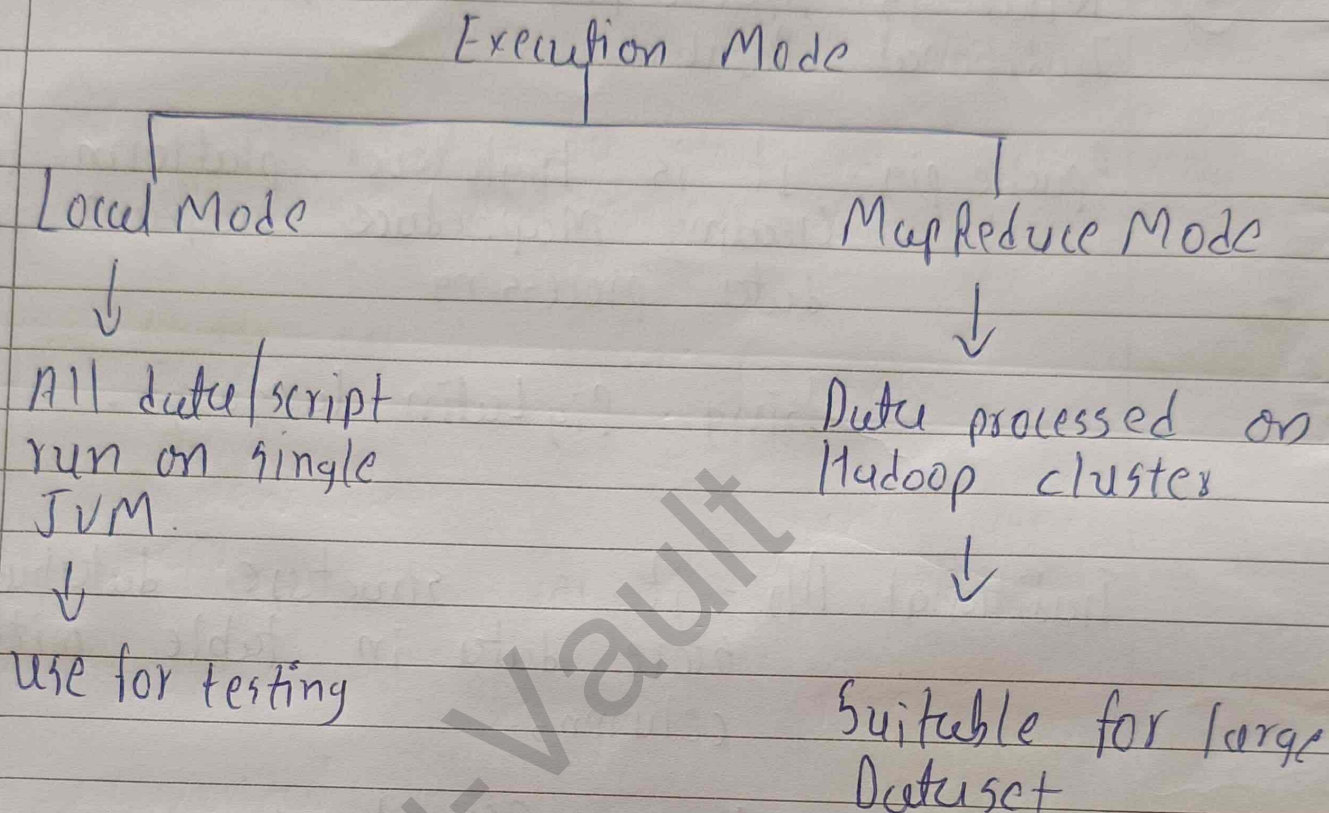
- Using Pig Latin language.

Traditional DB:- it is structure database that store data in table with row, columns.

- Using SQL Language.

feature	Apache Pig	Traditional DB
Language	Pig Latin	SQL
Schema	Optional	strict
Processing	Batch (Map Reduce)	Transactional
Storage	HDFS	RDBMS
Use case	Big Data	structure data

⑥ Execution Mode of pig & components



• Basic Components:-

Gruntshell

Pig Latin

Execution engine

UDFs (User define function)

7) Components of text Analytics pipeline

- Text collection:- Collect unstructured data from email, social media.
- Text Preprocessing:- Stop word removal
- Feature Extraction:- using Word embeddings.
- Text clustering:- Making text Group.
- Visualisation & Insight Extraction:- using Chart, dashboard.

8) Hive:- it is A warehouse infrastructure on Hadoop. it convert SQL-like to MapReduce jobs.

Feature	Hive	Traditioned DB
Query	Hive QL	SQL
Schema	flexible	fixed
Processing	Batch (Map Reduce)	Real-time
Speed	slow	fast
Use case	BDA	structure Data.

//_

(9) MongoDB - Unstructured, semi-structured data handled by NOSQL system that called MongoDB.

SQL:- structured data stored in relational database that called SQL.

feature	MongoDB	SQL
Data model	Document Based JSON	Table-based
Schema	Dynamic	fixed
Scalability	horizontal	vertical
Query Lang.	MQL	SQL
Best for	Unstructured & semi-structured data	structured
Speed	faster for Unstructured slow for semi-structured	fast for structured
Ex.	{ "User-id": 1, "name": "Hars", "feedback": ["cute", "Handsome"] }	create Table user (user_id INT, name varchar(50));

10

RDBMS Data type:-	Description	MangobB Data type
INT	32-bit integer	Number int
BIGINT	64-bit integer	Number Long
VARCHAR/ TEXT	High-decimal	Number Decimal
DATE	Text value	Date
BOOLEAN	True/false	Boolean
ARRAY	Binary Data	Array

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Stages for Data Analysis for Decision Points.

1. Data Collecting

2. Data Cleaning

3. Data Exploration

4. Modeling

5. Interpretation (Result)

6. Deployment.

(12)

Key Concepts & components of HBase.

Key:- 1. Column-oriented NoSQL DB on Hadoop
2. Store large table in multiple machine

Components:-

1. HMaster:- for control
2. Region Server:- Manage read/write region
3. Zookeeper:- for manage server
4. Hfile:- Physical storage format
5. Memstore:- In memory store before flush to disk.

13. Hive :- it is A warehouse infrastructure on hadoop. it convert SQL-like to MapReduce job.

Scalability factors:-

1. high scalability
2. SQL-like interface
3. Batch processing
4. Schema on Read
5. Extensibility through UDFs:- User define funct.
6. Integration
7. Compatibility with BI Tools.

(14) Map Reduce Architecture:-

Map Reduce :- is a programming model for processing large data set with distributed algorithm on Hadoop.

Components:-

- Input Data
- splitting
- Mapping
- Shuffling
- Reducing

Input
data

Deer	Deer	River
Car	Car	River
Deer	Car	River

Splitting

Deer	Deer	River
Car	Car	River
Deer	Car	Deer

Mapping
with key value

Deer	Deer	1
Car	1	
River	1	

Car	1	
Car	1	
River	1	

Deer	1	
Car	1	
River	1	

Shuffling
~~sorting~~
the key value
is paired

Deer 1, 1	Deer - 2
Deer 1	Deer - 1
River 1, 1, 1	River - 3
Car 1, 1, 1	Car - 3

Reduce

MapReduce Architecture:-

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